

Week 11 - Weather and Forecasting: Practice Questions

EART23001 Atmospheric Physics and Weather

Workload guide. The shaded question headings show the intended workload:

- **IN CLASS:** selected anchor problems that we may work through together in the timetabled session.
- **YOU SHOULD TRY:** expected independent practice after the session.
- **EXTRA CHALLENGE:** optional problems for students who want additional depth or a harder application.

Attempt the questions before looking at the worked answers. Show equations, substitutions and units for numerical work, and use sketches where they help. You are **not** expected to complete every Extra Challenge problem. A course-wide Symbols, Constants and Units sheet is provided separately; non-standard symbols are also defined locally.

YOU SHOULD TRY Question 1: Identify the air mass from its trajectory

For each trajectory reaching the British Isles, identify the most likely air mass and predict one characteristic weather feature after modification along the path.

1. Air travels northeast from the subtropical North Atlantic near the Azores, remaining over ocean for several days.
2. Air arrives from Greenland and the high-latitude North Atlantic after a long ocean crossing.
3. Air moves north from North Africa across France during summer.
4. In winter, very cold continental air moves west from central/eastern Europe, then crosses the North Sea before reaching eastern Britain.

YOU SHOULD TRY Question 2: Identify the front from the observed change

Name the frontal type represented by each idealised description.

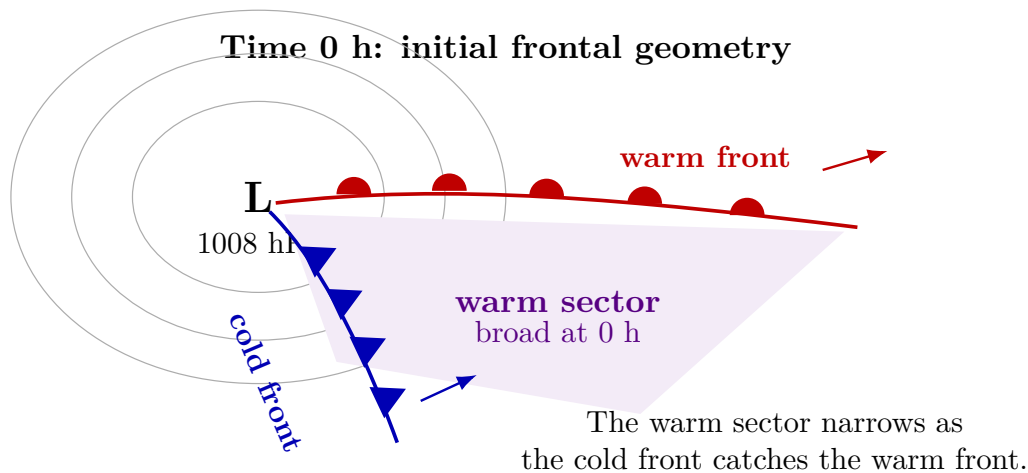
1. Surface air changes from 5°C to 11°C as a broad zone of layered cloud and steady rain passes.
2. Surface air changes from 12°C to 6°C as a narrow band of heavy showers and gusty winds passes.
3. A strong temperature contrast remains almost stationary over the same region for 18 hours, with persistent cloud and rain.
4. A faster cold front catches a warm front so that the warm sector is lifted away from the surface.

For cases 1 and 2, state whether the warmer air is replacing colder air or vice versa.

EXTRA CHALLENGE Question 3: Use pressure evolution to identify cyclone development and decay

An idealised mid-latitude cyclone has the following central pressure and surface warm-sector width:

Time (h)	0	12	24	36	48
Central pressure (hPa)	1008	996	984	988	997
Warm-sector width (km)	600	450	250	0	0



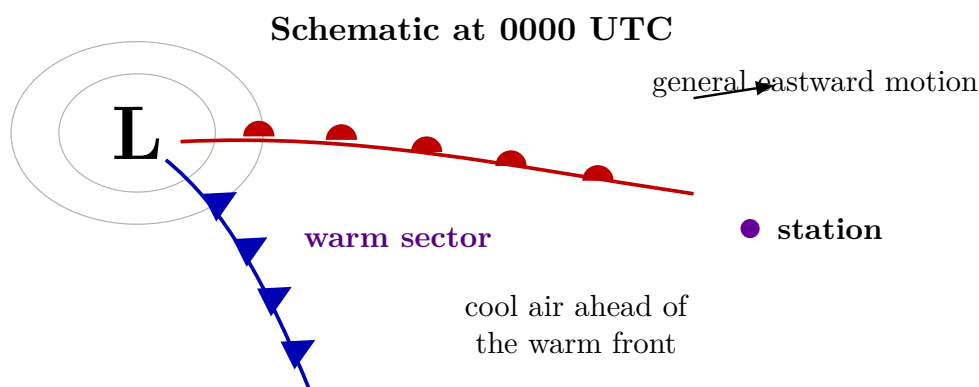
1. During which interval is the cyclone deepening most clearly?
2. When is the cyclone strongest according to central pressure?
3. At what time has surface occlusion clearly occurred according to the table?
4. What evidence indicates that the cyclone is filling/decaying after its mature stage?

IN CLASS Question 4: Diagnose a frontal passage from station observations

A station records the following sequence:

Time (UTC)	p (hPa)	T (°C)	Wind	Cloud/weather
0000	1008	5	ESE	high cloud increasing
0300	1004	6	SE	cloud thickening
0600	999	8	S	steady rain
0900	996	11	SW	rain easing; low cloud
1200	994	12	SW	mild, cloudy, drizzle
1500	997	8	W	brief heavy showers, gusty
1800	1003	6	NW	showers clearing

The schematic below shows the broad synoptic setting at 0000 UTC. It is not intended to give the frontal passage times directly; use the station observations above to diagnose those.



1. At approximately what time did the warm front pass?
2. At approximately what time did the cold front pass?
3. For each front, quote at least three pieces of observational evidence from the table.
4. During which interval was the station most clearly in the warm sector?

EXTRA CHALLENGE Question 5: Use isobar spacing on a synoptic chart

A surface analysis has 4 hPa isobar intervals. Near Region A, adjacent isobars are typically 80 km apart; near Region B they are 200 km apart. Assume similar latitude and air density.

1. Which region has the larger horizontal pressure gradient?
2. By what factor is it larger?
3. If the upper-air geostrophic wind in Region B is about 12 m s^{-1} , estimate the corresponding value in Region A under the same assumptions.
4. Why should a forecaster not expect the observed 10-m wind to equal this geostrophic estimate exactly?

YOU SHOULD TRY Question 6: Put the NWP forecast cycle in order

The following stages are deliberately scrambled:

- compare the forecast with later observations;
- collect surface, radiosonde, aircraft, radar and satellite observations;
- numerically integrate the atmospheric governing equations forward in time;
- combine observations with a previous forecast using data assimilation;
- produce the best estimate of the three-dimensional initial atmospheric state;
- represent unresolved processes such as some cloud/turbulence/surface exchange using parameterisations.

Put the stages into a sensible forecast-cycle order. Then identify two distinct sources of forecast error.

YOU SHOULD TRY Question 7: Interpret an ensemble forecast probabilistically

A 20-member ensemble forecast for Manchester gives the following outcomes for tomorrow afternoon:

- 14 members produce measurable rain;
 - 6 members produce more than 10 mm of rain;
 - 4 members remain completely dry.
1. Estimate the ensemble probability of measurable rain.
 2. Estimate the ensemble probability of more than 10 mm.
 3. Estimate the probability of a completely dry outcome.
 4. A single deterministic run predicts 12 mm. Explain why it would be misleading to report “12 mm will fall” without the ensemble information.

EXTRA CHALLENGE Question 8: Diagnose a blocking-high scenario

For four consecutive days, analysis charts show a 1035 hPa anticyclone remaining almost stationary over western Europe. Atlantic cyclones that would normally move east are instead diverted north around it.

1. Why is this described as a blocking high?
2. What broad surface-wind and vertical-motion pattern is expected near the centre of the anticyclone?

3. Give one likely summer weather consequence beneath the block and one possible winter consequence.
4. Why can regions around the edge of a block experience persistent wet weather even while the centre is dry?

EXTRA CHALLENGE Question 9: AI and physics-based forecasting

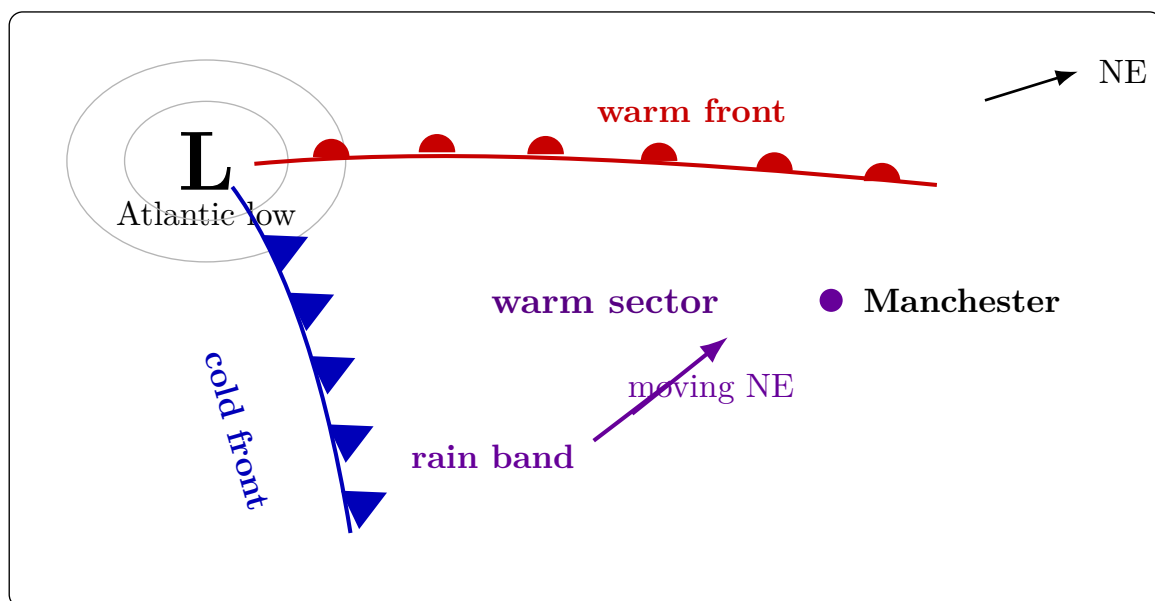
A conventional NWP model and a data-driven machine-learning model both produce a 5-day forecast. The ML model runs much faster, while the NWP model explicitly evolves conservation laws and parameterised physical processes.

1. Give one scientific or operational advantage of the ML approach.
2. Give one advantage of retaining physics-based NWP.
3. Explain why a forecasting centre might use both rather than regard them as mutually exclusive replacements.

IN CLASS Question 10: Integrative UK weather case: chart, sounding and radar

At 0900 UTC a deep Atlantic low lies west of Ireland. A warm front is moving northeast across Wales and a cold front is approaching southwest England. Manchester lies in the warm sector. Surface observations at Manchester are $T = 11^\circ\text{C}$, dew point 9°C , pressure 995 hPa falling at 2 hPa in 3 h, and wind 210° at 12 m s^{-1} . A nearby sounding shows a nearly saturated layer from 1–3 km, a freezing level near 2.2 km and winds increasing to 30 m s^{-1} by 5 km. Radar shows a narrow band of heavier precipitation 120 km to the southwest moving northeast at 50 km h^{-1} .

Schematic synoptic situation at 0900 UTC



1. Estimate when the radar rain band will reach Manchester if its speed remains constant.
2. Use the surface temperature/dew-point pair and sounding to assess whether cloud is likely before frontal passage.
3. Explain why the falling surface pressure and strengthening winds aloft are dynamically consistent with an approaching cyclone/frontal system.
4. What precipitation phase is most likely at the surface, and why?
5. Predict the likely changes in wind, temperature and pressure as the cold front passes.
6. Identify one important uncertainty in this six-hour forecast that cannot be removed by the information supplied.